DO NOT WRITE IN THIS MARGIN
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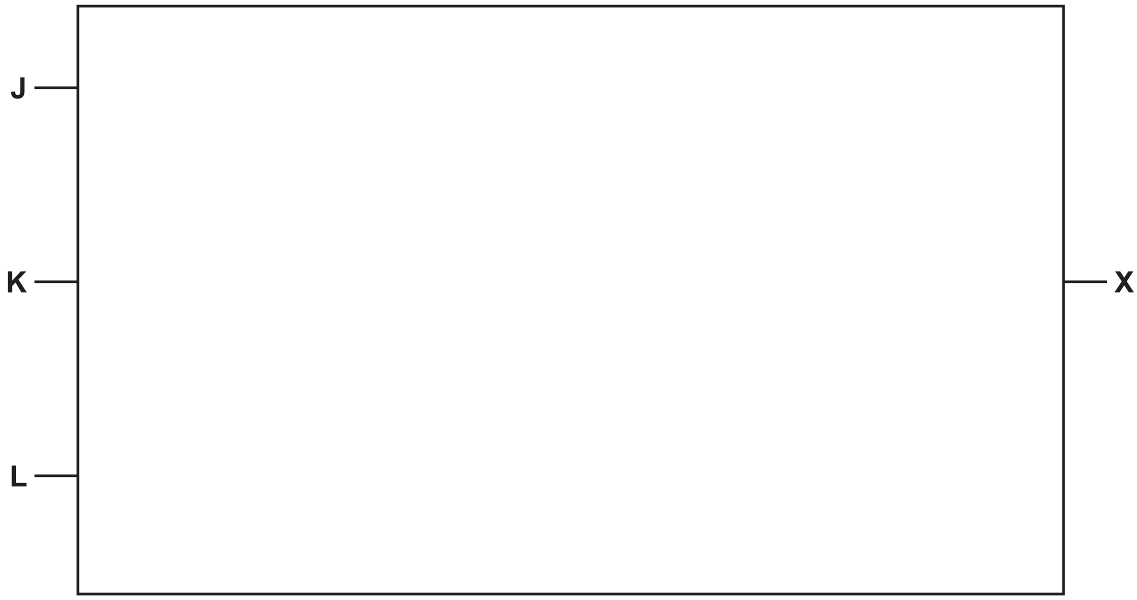
**6** Consider the logic expression:

**X** = (**J** XOR NOT **K**) NAND NOT **L**

**(a)** Draw a logic circuit for the logic expression.
Each logic gate must have a maximum of **two** inputs.
Do **not** simplify the logic expression.




[4]

**(b)** Complete the truth table for the given logic expression.

| J | K | L | Working space | X |
|---|---|---|---|---|
| 0 | 0 | 0 | | |
| 0 | 0 | 1 | | |
| 0 | 1 | 0 | | |
| 0 | 1 | 1 | | |
| 1 | 0 | 0 | | |
| 1 | 0 | 1 | | |
| 1 | 1 | 0 | | |
| 1 | 1 | 1 | | |

[4]